# Mahek Sota

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## Experience

**Software Engineer (Backend & AI)**, University of Mississippi — Jul 2024 – Present

- **Product Ownership:** Lead end-to-end development of production research systems, translating faculty and administrator needs into applications supporting **10K+ students and 100+ faculty**.
- **Applied AI:** Built an LLM-assisted citation pipeline using semantic analysis and regex, achieving **95%+ extraction accuracy**, **98% classification precision**, and reducing review time from **30 to 2 minutes**.
- **Search and Retrieval:** Built an agentic RAG assistant using FAISS and local LLM verification to address librarian and administrator search bottlenecks, improving retrieval speed by **60%** and reducing manual review by **70%**.
- **Data Infrastructure:** Designed and operated Python, PostgreSQL, and Kafka pipelines processing **50M+ records**, implementing batching, delta synchronization, failure handling, and observability to reduce ingestion latency by **30%**.
- **Platform Engineering:** Modernized legacy services into containerized **Go microservices on AWS EKS**, implementing GitHub Actions CI/CD, centralized logging, JWT authentication, and Redis-backed rate limiting.
- **Research Intelligence:** Built and deployed research discovery and resource analytics platforms integrating OpenAlex and Dimensions, enabling subsecond search across **6,900+ researcher profiles** and **44.5K+ publications** while supporting evidence-based collection decisions.

**Software Engineer Intern**, Stealth Startup — Jun 2023 – Aug 2023

- Built a **React and FastAPI geospatial application** visualizing **10K+ data points** across 3+ interactive views.
- Integrated external space datasets through **5+ REST APIs** for ingestion, transformation, and retrieval.
- Added caching, validation, and error handling with **automated 30-second refreshes** for reliable real-time updates.

**Course Producer**, USC Viterbi School of Engineering — Jan 2023 – May 2023

- Mentored **100+ students** in SQL, ER modeling, and relational database design through weekly labs and office hours; created **15+ practice problems and real-world database case studies**.

**Machine Learning Intern**, Friends for Inclusion — Dec 2021 – Jan 2022

- Built an AI-assisted audiobook translation tool for visually impaired users, optimizing Hugging Face Transformer pipelines to achieve **80% accuracy**.

**Software Engineer Intern**, Ashtech — Jun 2021 – Aug 2021

- Served as a **client-facing engineer**, translating business requirements into feasible product features.
- Built accessible web and PayPal payment workflows, reducing latency by **40%** through lazy loading and modularization; added automated tests using pytest and Jest.

## Projects

- **CI/CD Observability Platform** – Built a real-time dashboard to track CI/CD pipelines, container health, and resource utilization; integrated Prometheus and Grafana with alerting to improve deployment visibility and reduce troubleshooting time. *Tech Stack: React, Spring Boot, PostgreSQL, Prometheus, Grafana, Docker, Kubernetes*
- **Cross-Border Care Coordination Platform** – Built a full-stack platform supporting provider comparison, medical-record intake, and case tracking; implemented asynchronous document processing, role-based access for patients and coordinators, and search over a seeded dataset of **500+ synthetic provider records**. *Tech Stack: React, FastAPI, PostgreSQL, Redis, Celery, Docker, AWS*
- **AI Agent Platform with MCP Integrations** – Built an agentic AI platform integrating **3 workflow categories** and **5+ MCP tools** across multiple data sources; implemented Redis caching and OpenTelemetry tracing for reliable tool execution and end-to-end observability. *Tech Stack: MCP, FastAPI, PostgreSQL, Redis, Docker, OpenTelemetry*

## Education

**University of Southern California**, Los Angeles, CA — *Aug 2022 – May 2024*
Master of Science in Computer Science **(GPA: 3.5/4.0)**

**Mumbai University, Thadomal Shahani Engineering College**, Mumbai, India — *Aug 2018 – May 2022*
Bachelor of Technology in Computer Science **(GPA: 9.5/10.0)**

## Technical Skills

**Languages**: Python, Java, C++, JavaScript, SQL, Go, TypeScript
**Frontend**: React, Next.js, TypeScript, Tailwind CSS, React Native
**Backend**: Kafka, Redis, FastAPI, Spring Boot, Django, Flask, Node.js, REST APIs, GraphQL, gRPC, OAuth2, JWT
**Databases**: PostgreSQL, MongoDB, Redis, FAISS
**DevOps & Cloud**: Docker, Kubernetes, Terraform, GitHub Actions, Jenkins, AWS, GCP, Prometheus
**Other**: Git, Agile/Scrum, JIRA